CS & IT ENGINEERING

Discrete Mathematics

Graph Theory

Lecture No.- 07

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Recap of Previous Lecture

Topic Types of Graph Part-03

Topics to be Covered

Topic Types of Graph Part-04

Topic : Types of Graph

Q: Consider a Graph vertices are represented as n-bit signal. Two vertices are connected with each other, when there bit differences changes by 1-bit, what will be total edges in the Graph. ($n \geq 3$).

Topic : Types of Graph

Graph

vertices $\Rightarrow$ n-bit signal

$n = 2$ (2-bit signal)

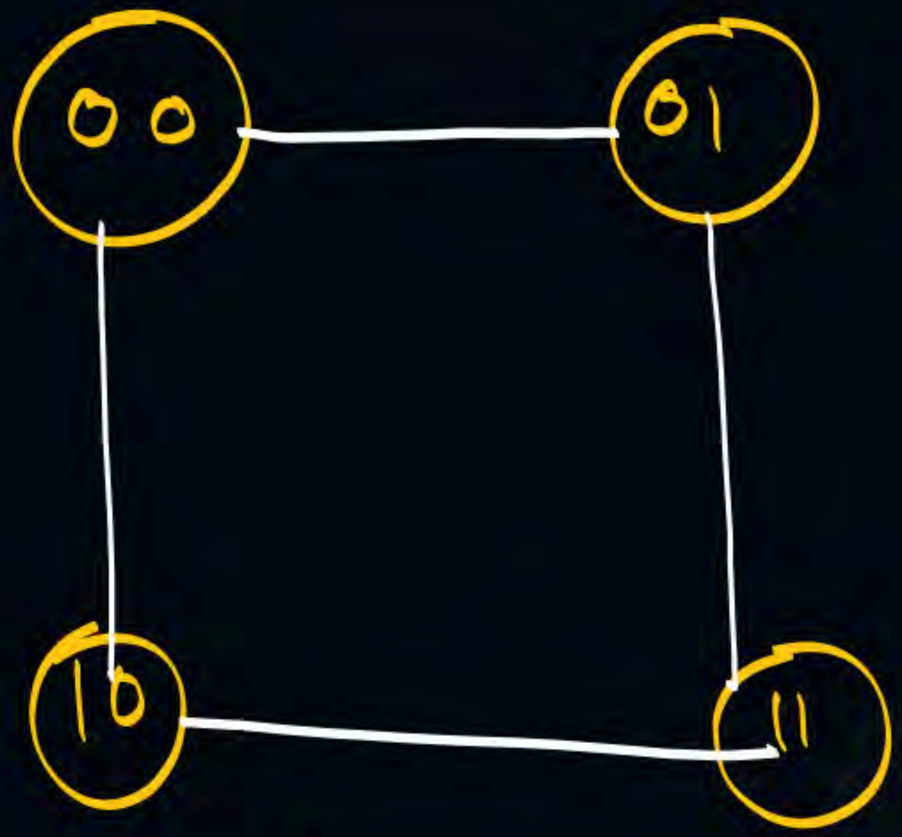
no. of codes we can generate $= 2^2$

$v_1 \rightarrow 00$

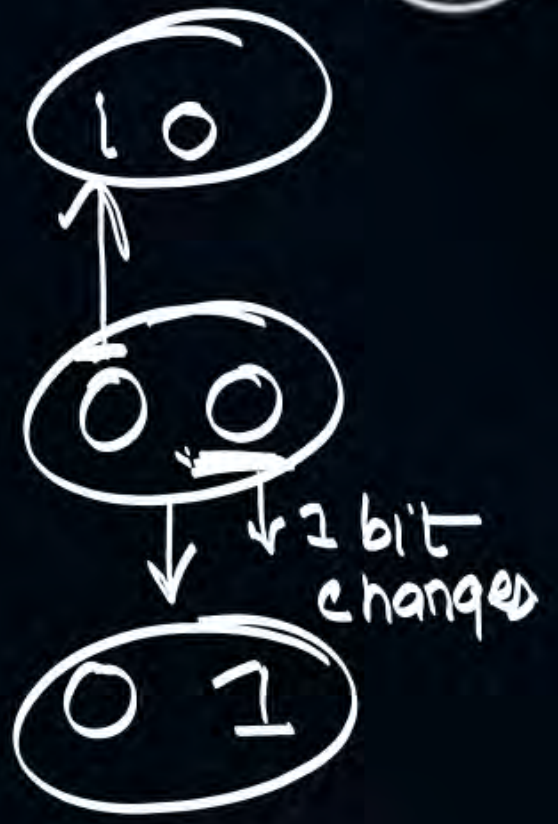
$v_2 \rightarrow 01$

$v_3 \rightarrow 10$

$v_4 \rightarrow 11$



Degree of each vertex is 2.

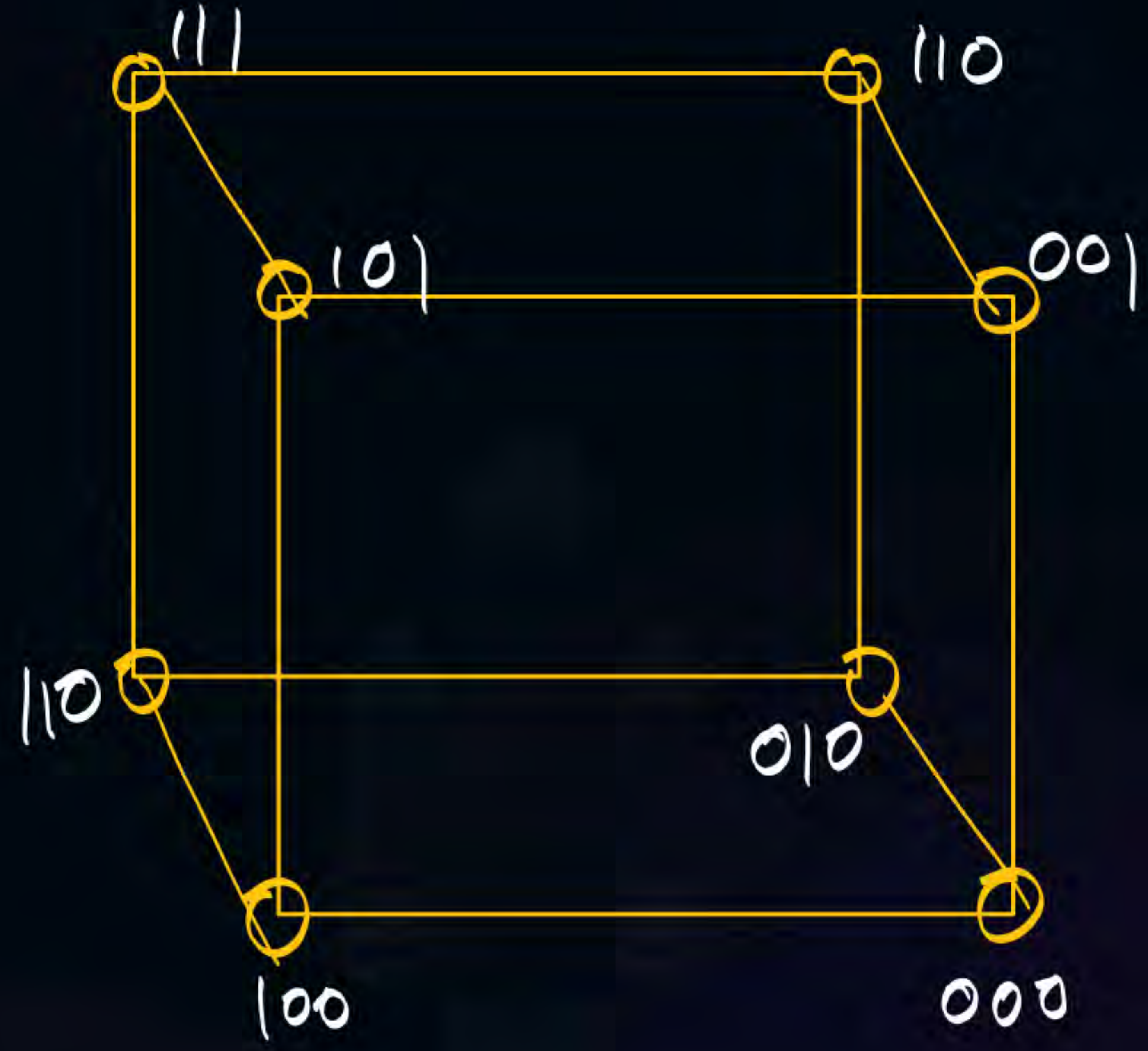


Topic : Types of Graph

$n = 3$ bit

$$2^3 \left\{ \begin{array}{c} 000 \\ 001 \\ 010 \\ 011 \\ 100 \\ \vdots \\ 111 \end{array} \right.$$

Total vertices = 2^3
Degree of each vertex is 3.



Topic : Types of Graph

Total vertices $= 2^n$ (n-bit signal)

degree of each vertex is n.

$$\Sigma d(v_i) = 2e$$

$$2^n \cdot n = 2e$$

$$e = \frac{2^n \cdot n}{2}$$

$$\boxed{e = n \cdot 2^{n-1}}$$

$e(G) + e(\bar{G}) = K_n.X$ (n-bit signal)

$e(G) + e(\bar{G}) = K_V$ (Total vertices $= V = 2^n$)

$$e(G) + e(\bar{G}) = \frac{V(V-1)}{2}$$

$$n \cdot 2^{n-1} + e(\bar{G}) = \frac{2^n(2^n - 1)}{2} \quad (V = 2^n)$$

$$e(\bar{G}) = \frac{2^n(2^n-1)}{2} - n \cdot 2^{n-1}$$

Topic : Types of Graph

$e(Q_n) = n \cdot 2^{n-1}$

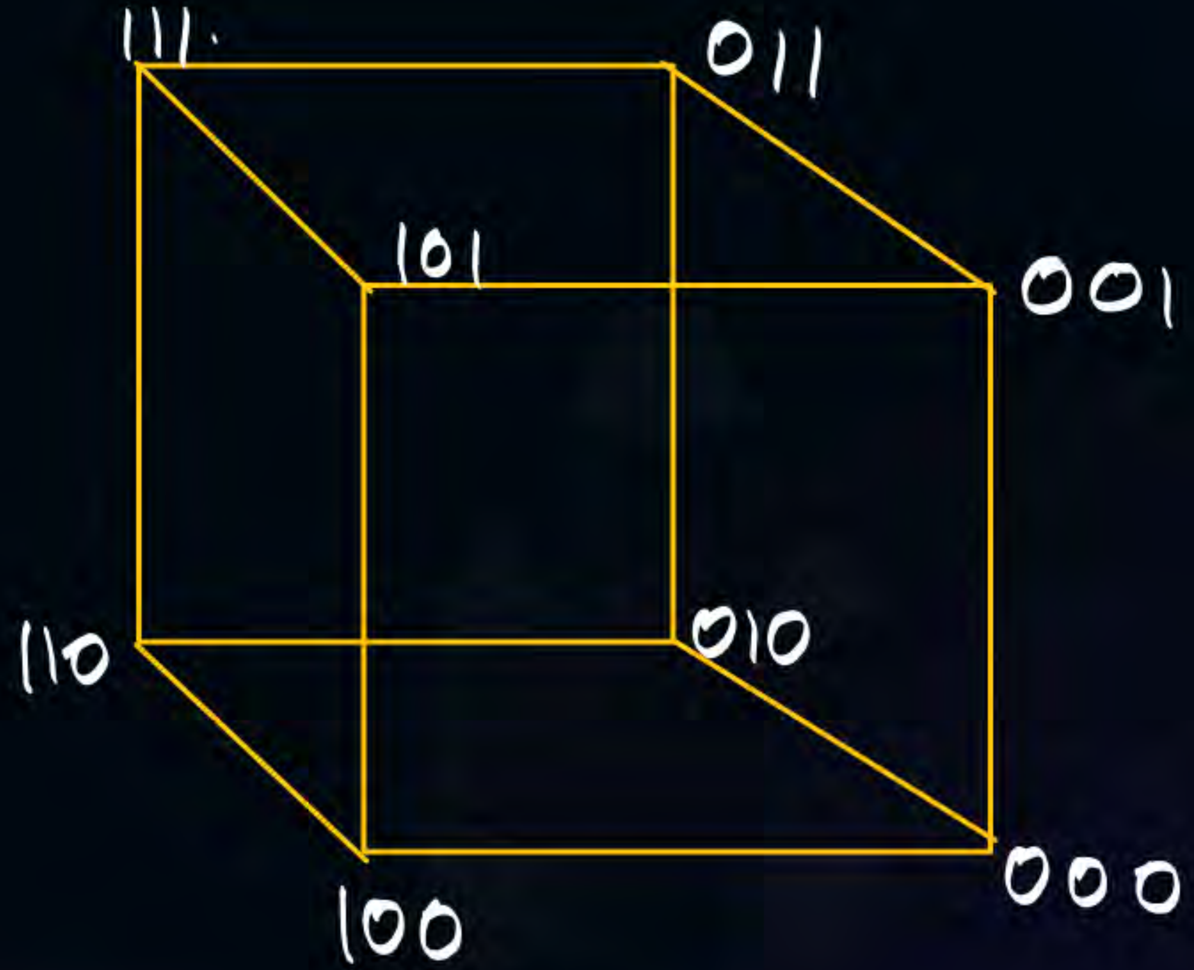
Hypercube (Q_n) ↳ (n-bit signal)

Q_1

Total vertices $= 2^1$
Degree of each vertex is 1.

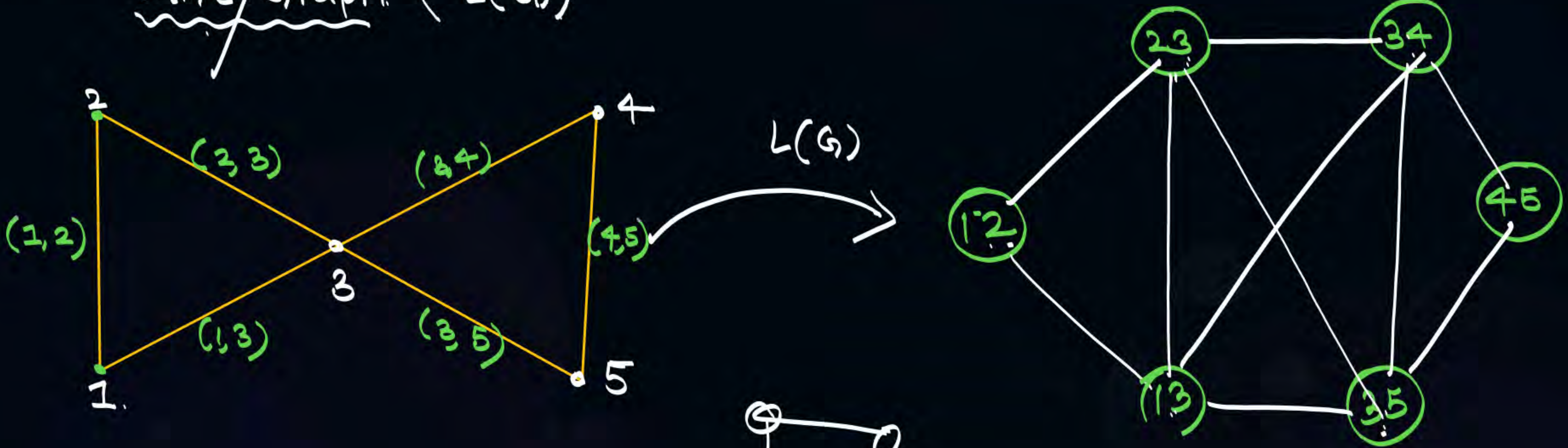
Q_2

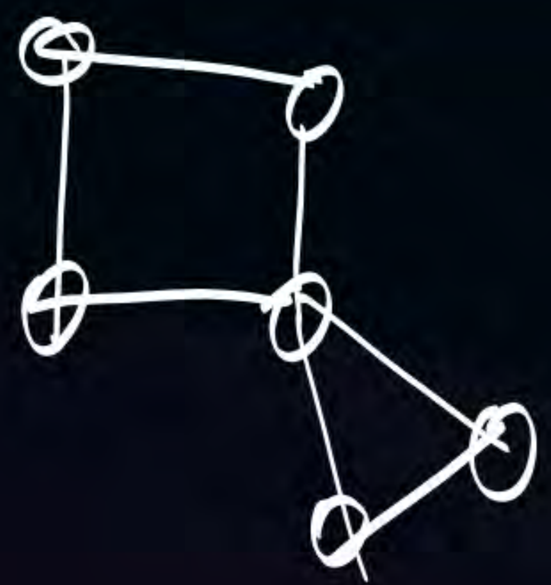
Q_3



Topic : Types of Graph

Line Graph (L(G))





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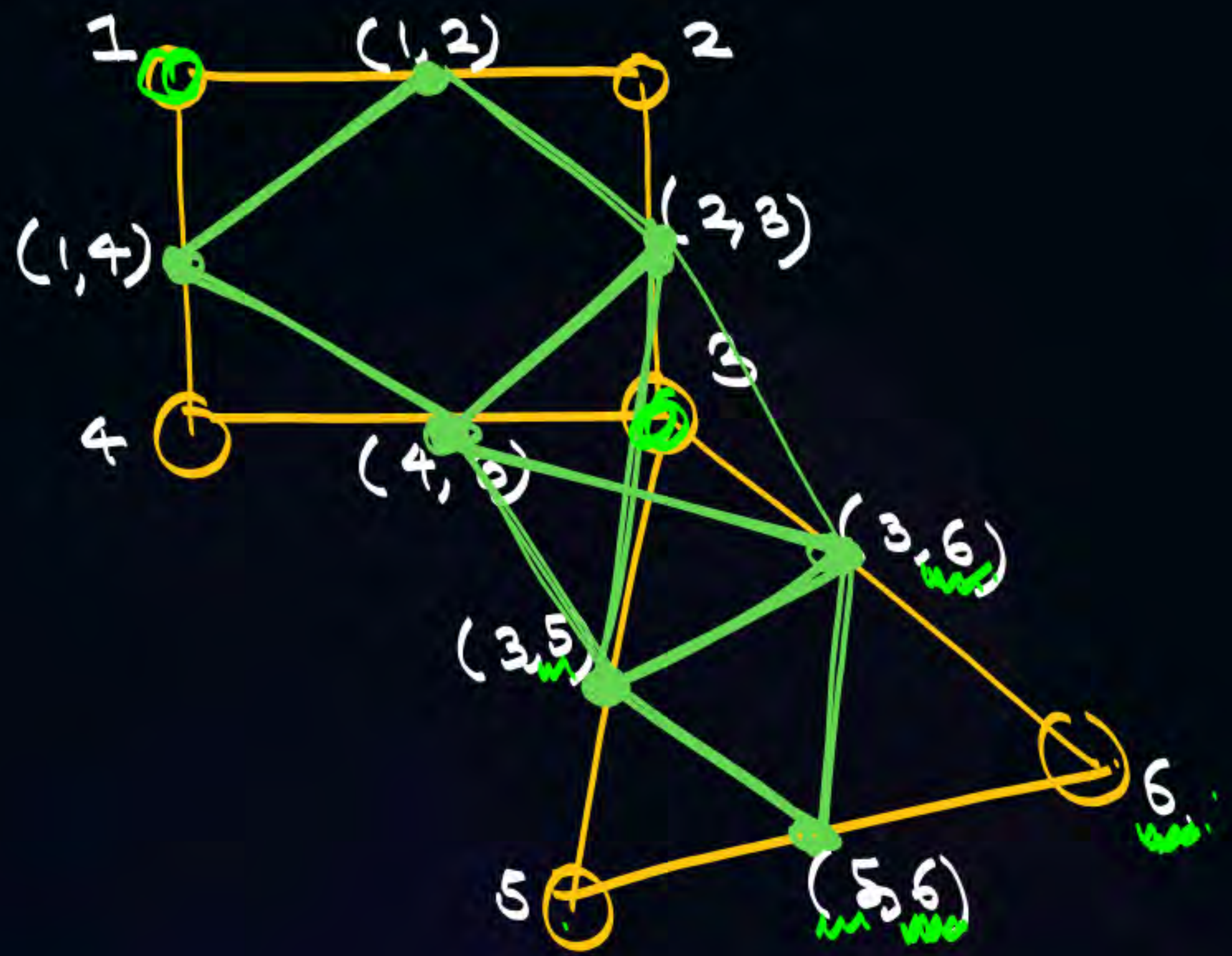
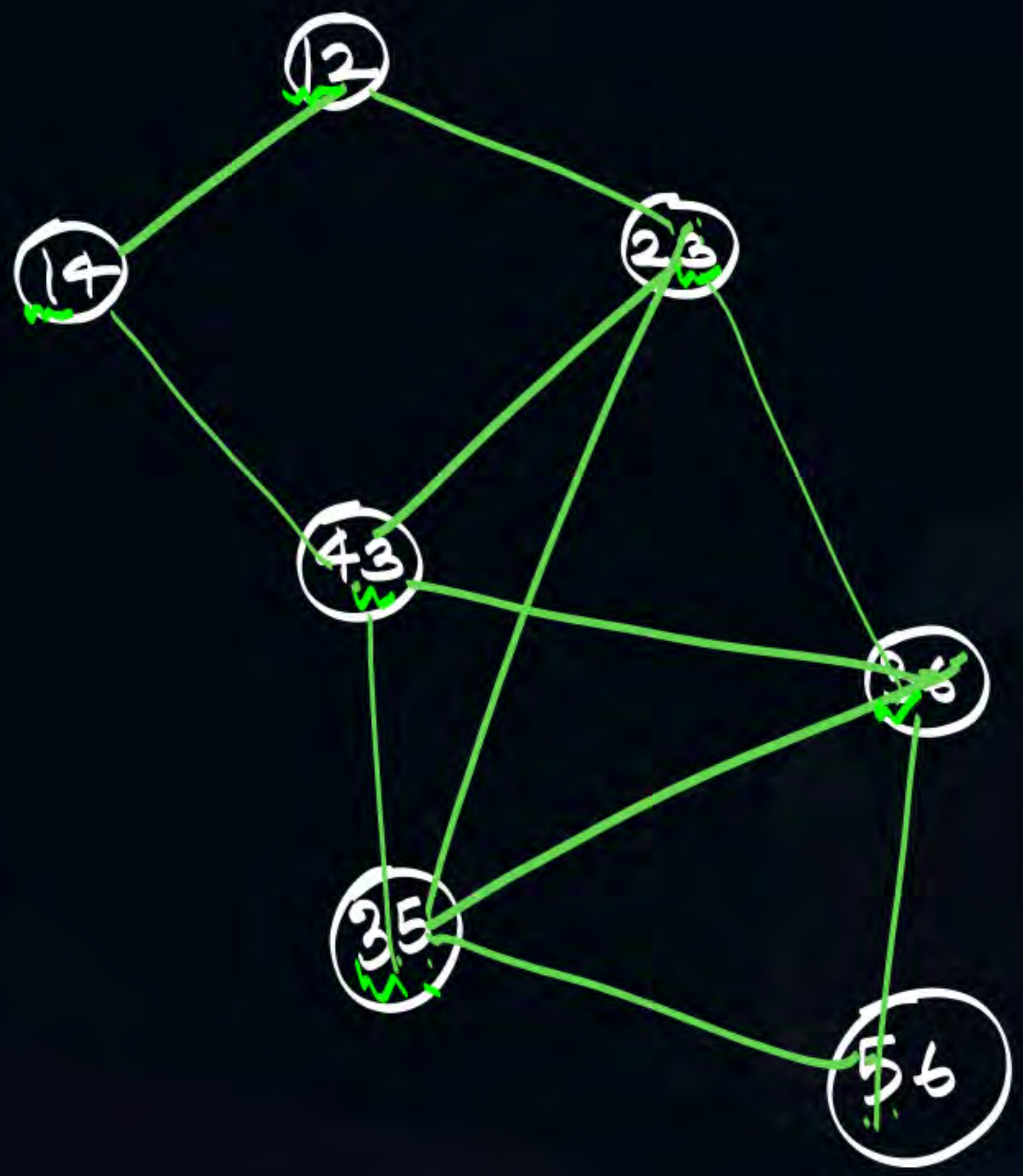




Topic : Types of Graph

Degree of each vertex of $L(K_n)$ will be ____.

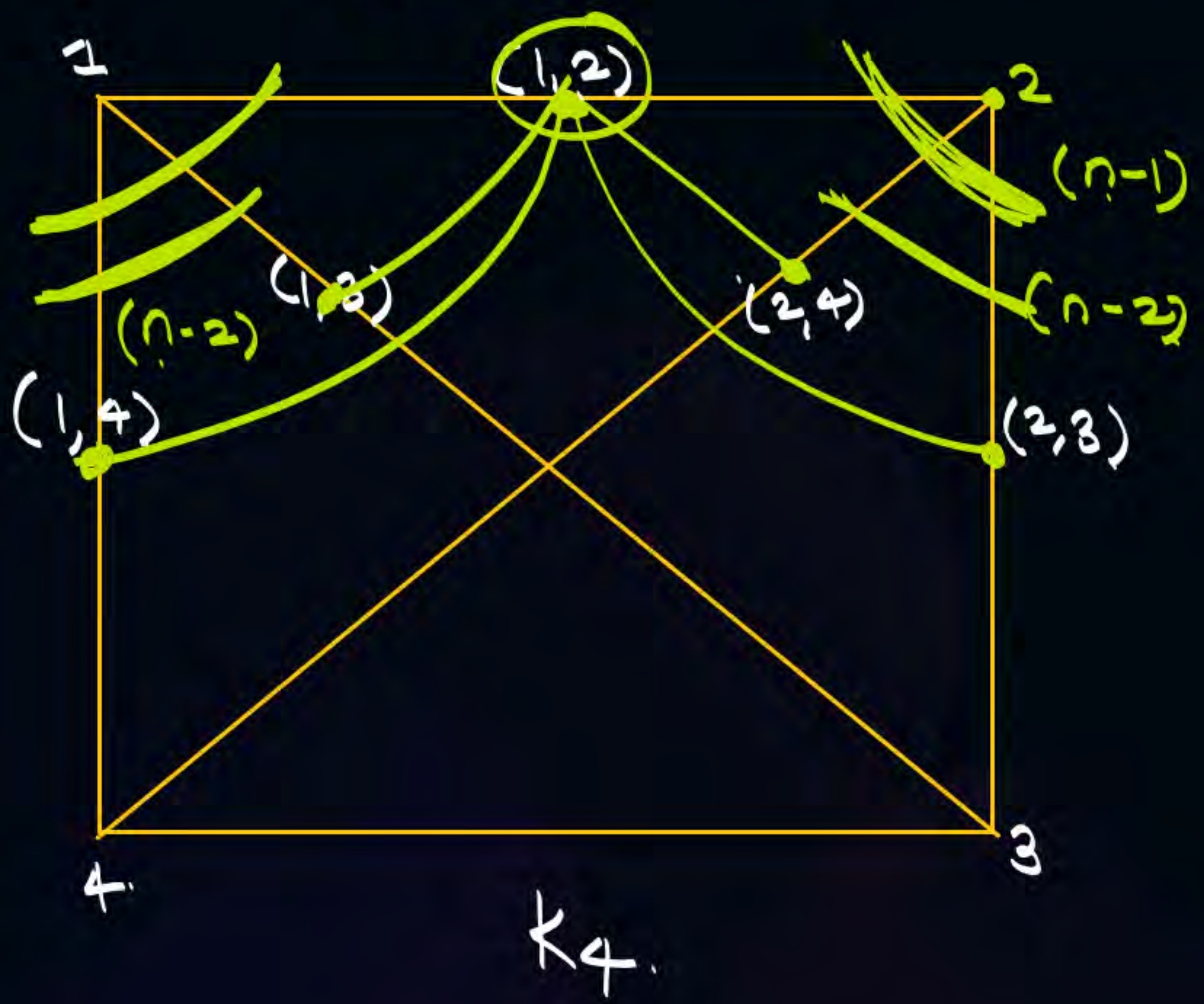
a) $n-1$

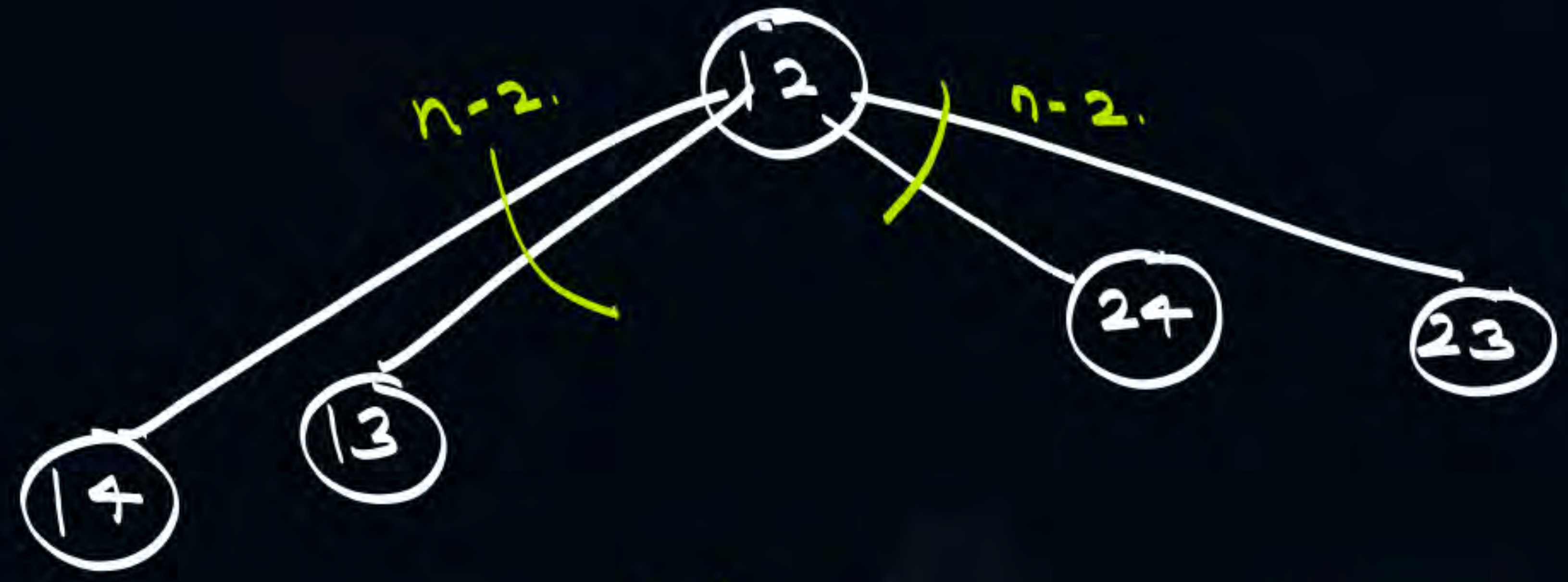
b) $2(n-1)$

c) $2(n-2)$

d) $n-2$

Topic : Types of Graph





$2(n-2)$

2 mins Summary

Topic	One
Topic	Two
Topic	Three
Topic	Four
Topic	Five

Types of Graph Part-04

THANK - YOU